

## EXERCISES

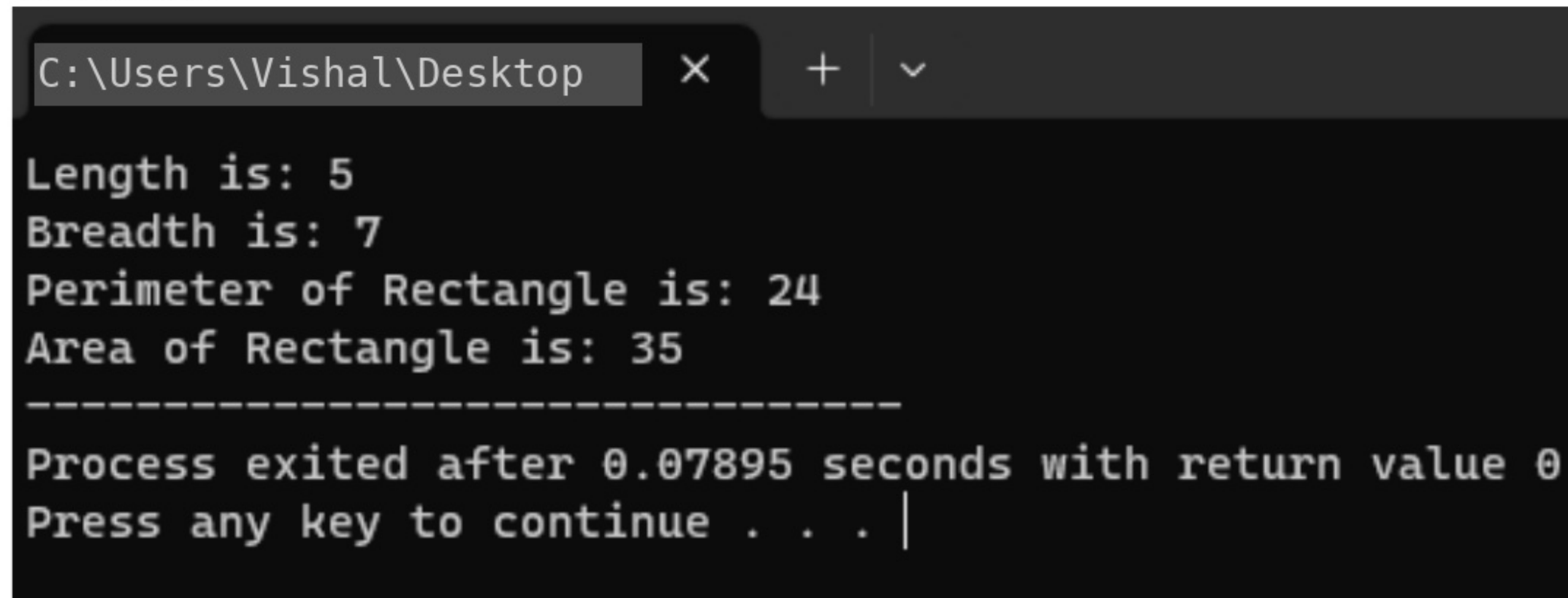
### QUESTION 1:

Length and breadth of a rectangle are 5 and 7 respectively. Write a program to calculate the area and perimeter of the rectangle.

### CODE:

```
#include<iostream>
using namespace std;
int main(){
    int l(5),b(7);
    int p=(l+b)*2;
    int area=l*b;
    cout<<"Length is: "<<l<<endl
         <<"Breadth is: "<<b<<endl;
    cout<<"Perimeter of Rectangle is: "<<p<<endl
         <<"Area of Rectangle is: "<<area;
}
```

### OUTPUT:



```
C:\Users\Vishal\Desktop  X  +  v

Length is: 5
Breadth is: 7
Perimeter of Rectangle is: 24
Area of Rectangle is: 35
-----
Process exited after 0.07895 seconds with return value 0
Press any key to continue . . . |
```

## QUESTION 2:

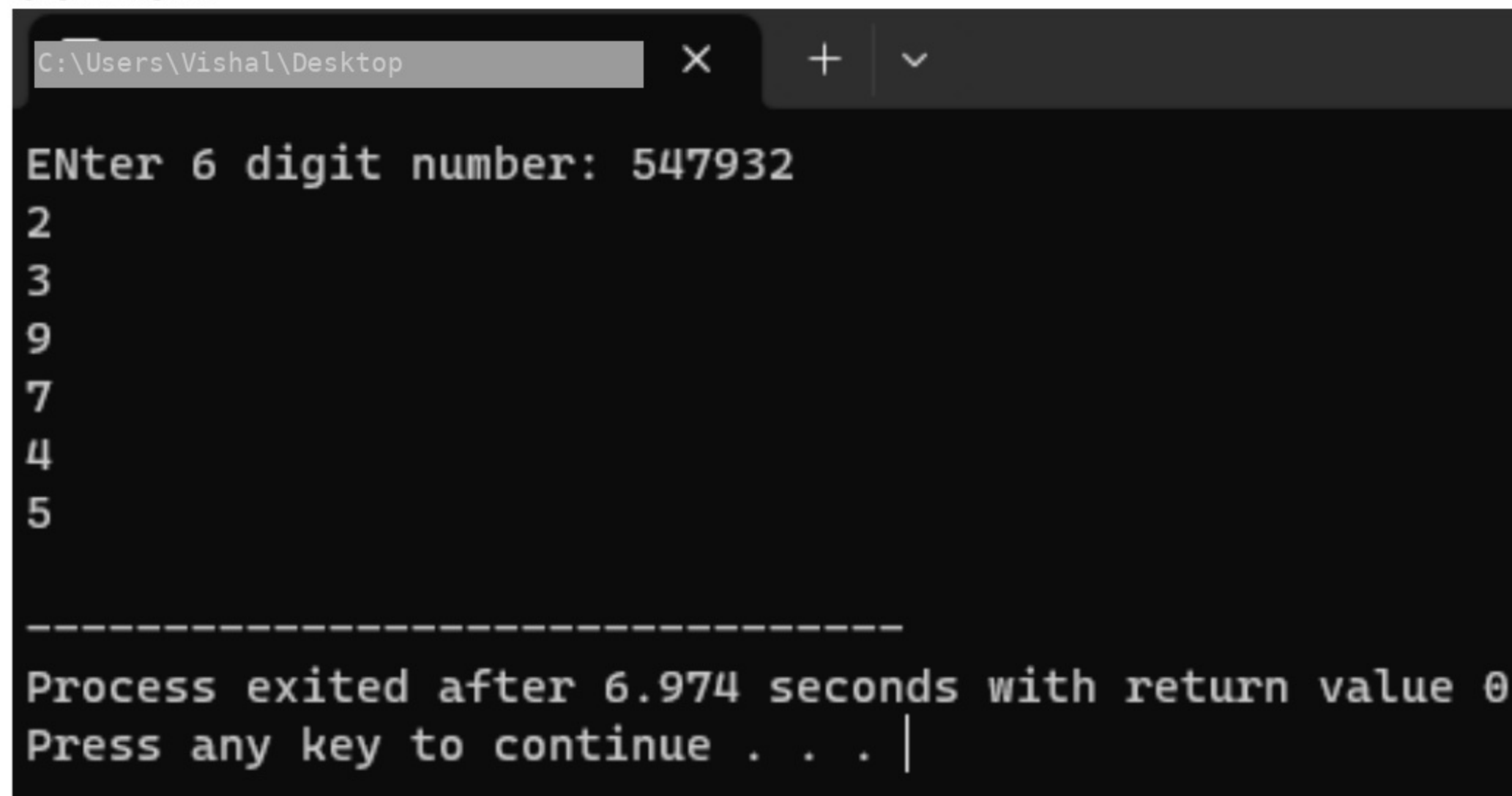
Write a program that inputs a six-digit integer, separates the integer into its individual digits and prints the digits vertically using modulus operator. For example, if the user types 953647, the program should print:

7  
4  
6  
3  
5  
9

## CODE:

```
#include <iostream>
using namespace std;
int main(){
    int num;
    cout<<"ENter 6 digit number: ";
    cin>>num;
    cout<<num%10<<endl;
    cout<<num/10%10<<endl;
    cout<<num/100%10<<endl;
    cout<<num/1000%10<<endl;
    cout<<num/10000%10<<endl;
    cout<<num/100000<<endl;
    return 0;
}
```

## OUTPUT:



```
C:\Users\Vishal\Desktop
Enter 6 digit number: 547932
2
3
9
7
4
5

-----
Process exited after 6.974 seconds with return value 0
Press any key to continue . . . |
```



### QUESTION 3:

Write a program to convert Fahrenheit into Celsius.

$$C = \frac{5}{9} (F - 32)$$

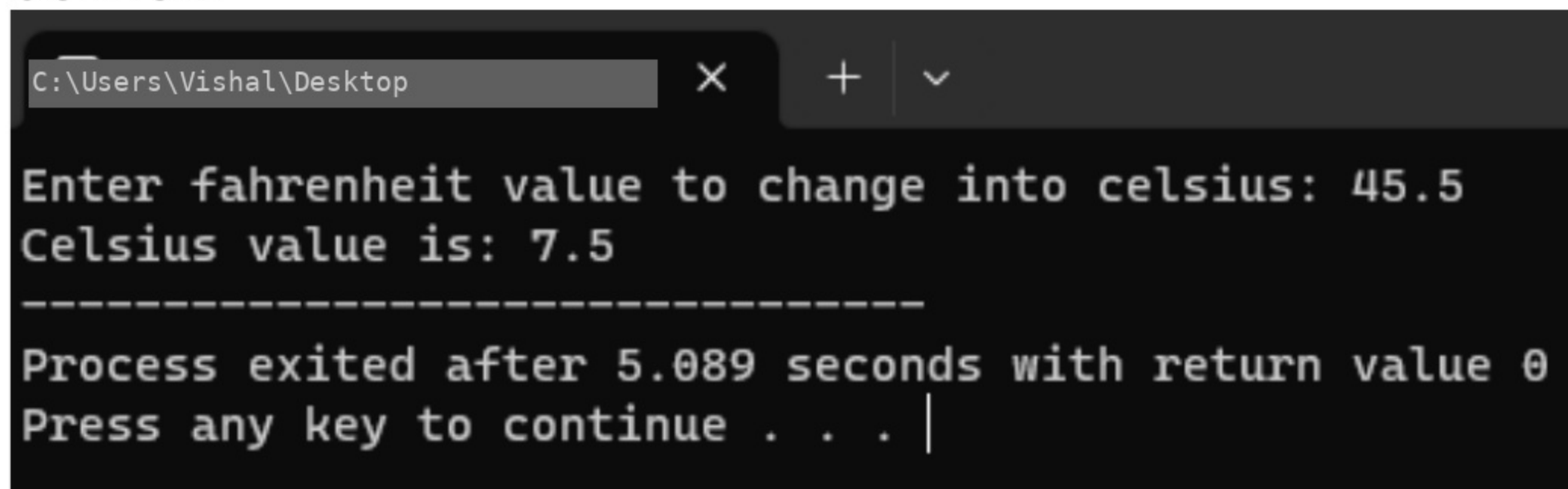
### CODE:

```
#include <iostream>
using namespace std;
int main(){

double f,c;
    cout<<"Enter fahrenheit value to change into celsius: ";
    cin>>f;
c=(double)5/9*(f-32);
    cout<<"Celsius value is: "<<c;

return 0;
}
```

### OUTPUT:

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Vishal\Desktop'. The window contains the following text: 'Enter fahrenheit value to change into celsius: 45.5', 'Celsius value is: 7.5', a separator line of dashes, and 'Process exited after 5.089 seconds with return value 0'. The prompt 'Press any key to continue . . . |' is visible at the bottom.

```
C:\Users\Vishal\Desktop
```

```
Enter fahrenheit value to change into celsius: 45.5
Celsius value is: 7.5
-----
Process exited after 5.089 seconds with return value 0
Press any key to continue . . . |
```

#### QUESTION 4:

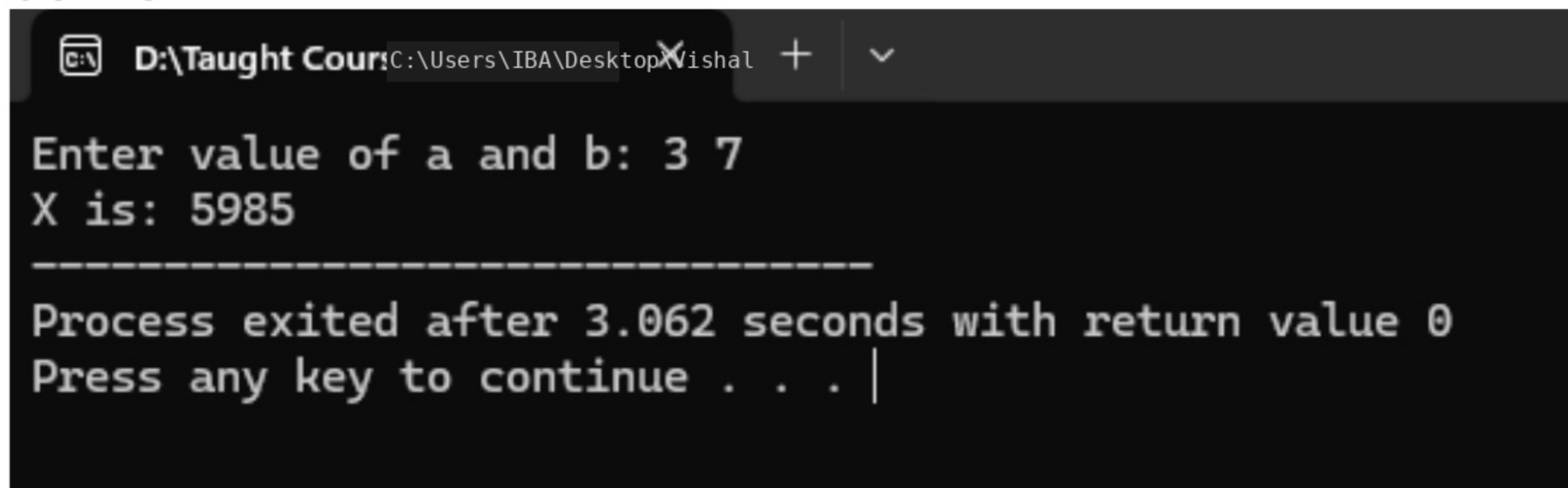
Write a program that finds the value of X by using given formula. Take value of 'a' and 'b' from user.

$$X = 2a^2b ( a + b^2 ) - 3a^3b$$

#### CODE:

```
#include<iostream>
using namespace std;
int main(){
    float a,b,X;
    cout<<"Enter value of a and b: ";
    cin>>a>>b;
    X=(2*a*a*b)*(a+b*b)-(3*a*a*a*b);
    cout<<"X is: "<<X;
    return 0;
}
```

#### OUTPUT:



```
D:\Taught Course\Users\IBA\Desktop\Xishal
Enter value of a and b: 3 7
X is: 5985
-----
Process exited after 3.062 seconds with return value 0
Press any key to continue . . . |
```



### QUESTION 5:

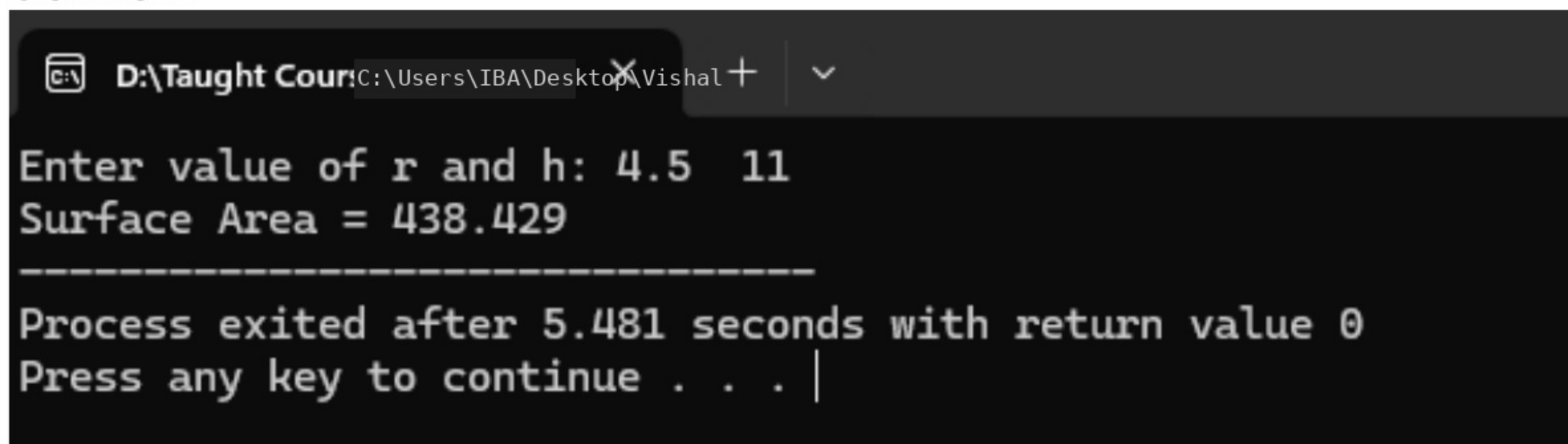
Create a program to calculate and display the surface of the cylinder given the radius (r) and height (h) using the formula:

$$\text{Surface\_Area} = 2\pi r^2 + 2\pi rh$$

### CODE:

```
#include<iostream>
using namespace std;
int main(){
    float Surface_Area, r, h, pi=22.0/7;
    cout<<"Enter value of r and h: ";
    cin>>r>>h;
    Surface_Area=2*pi*r*r + 2*pi*r*h;
    cout<<"Surface Area = "<<Surface_Area;
    return 0;
}
```

### OUTPUT:



The screenshot shows a Windows command prompt window with the title bar "D:\Taught Cour: C:\Users\IBA\Desktop\Visual +". The prompt displays the following text:

```
Enter value of r and h: 4.5 11
Surface Area = 438.429
-----
Process exited after 5.481 seconds with return value 0
Press any key to continue . . . |
```

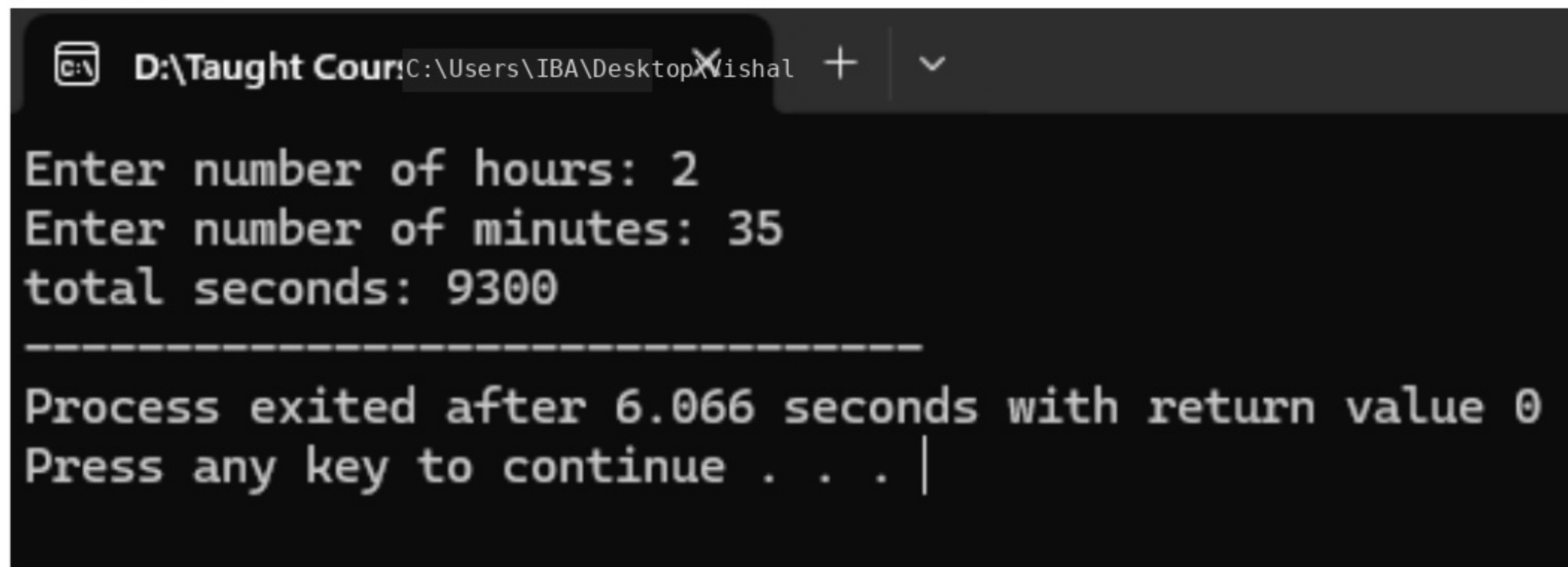
### QUESTION 6:

Write a program that takes two values for hours and minutes and then calculates total seconds

#### CODE:

```
#include<iostream>
using namespace std;
int main(){
    int hour,mint,sec;
    cout<<"Enter number of hours: ";
    cin>>hour;
    cout<<"Enter number of minutes: ";
    cin>>mint;
    sec=(mint*60)+(hour*60*60);
    cout<<"total seconds: "<<sec;
    return 0;
}
```

#### OUTPUT:



```
D:\Taught Cour... C:\Users\IBA\Desktop\Wishal + v
Enter number of hours: 2
Enter number of minutes: 35
total seconds: 9300
-----
Process exited after 6.066 seconds with return value 0
Press any key to continue . . . |
```



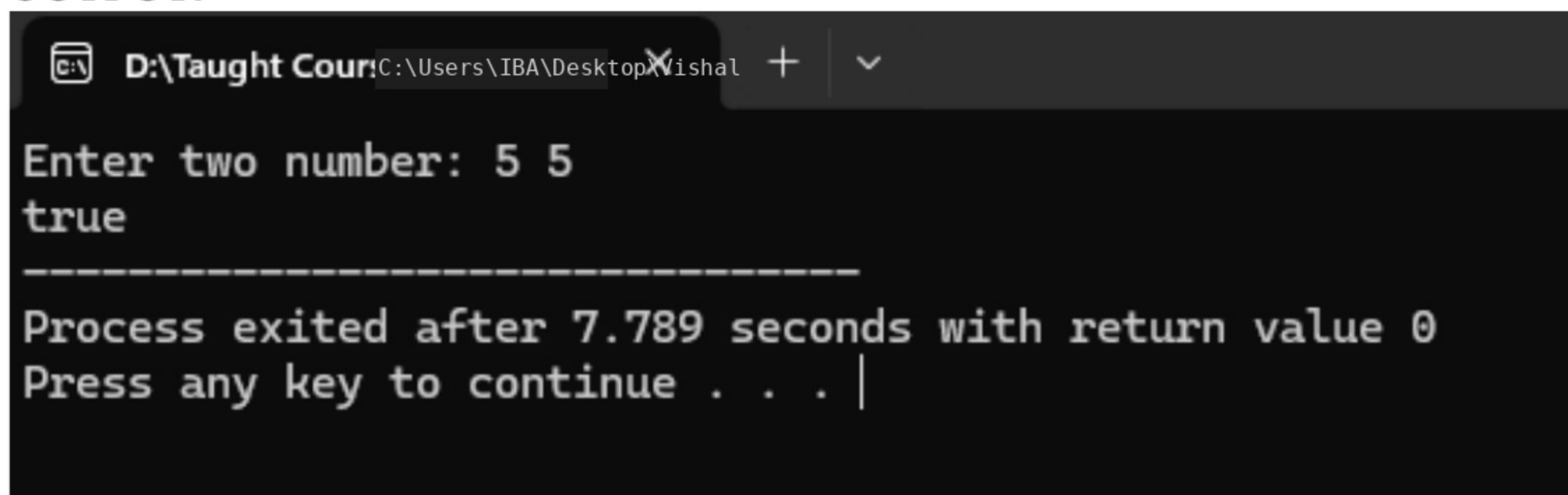
### QUESTION 7:

Write a program that takes two numbers from user and checks if the two numbers are equal or not by returning true or false respectively.

### CODE:

```
#include <iostream>
using namespace std;
int main(){
    int a,b;
    bool result;
    cout<<"Enter two number: ";
    cin>>a>>b;
    result=(a==b);
    cout<<"boolalpha"<<result;
    return 0;
}
```

### OUTPUT:

A screenshot of a terminal window showing the execution of a C++ program. The window title is "D:\Taught Cour: C:\Users\IBA\Desktop\X Vishal". The program prompts "Enter two number: 5 5" and outputs "true". A dashed line separates the output from the status message "Process exited after 7.789 seconds with return value 0". The prompt "Press any key to continue . . ." is followed by a vertical bar cursor.

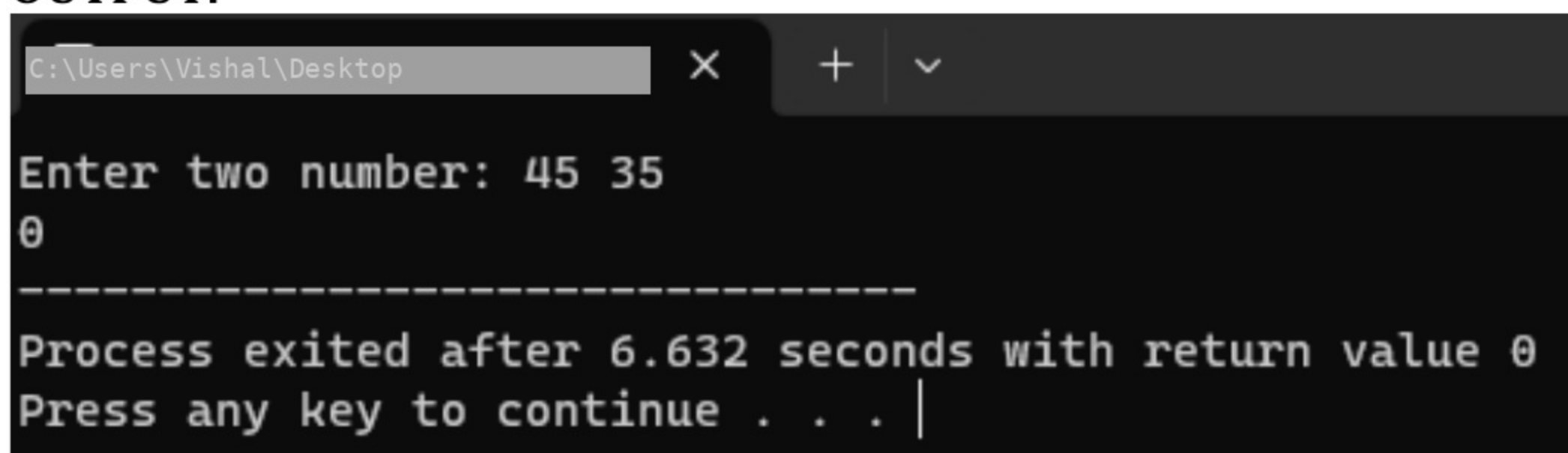
```
D:\Taught Cour: C:\Users\IBA\Desktop\X Vishal + v
Enter two number: 5 5
true
-----
Process exited after 7.789 seconds with return value 0
Press any key to continue . . . |
```

**QUESTION 8:**

Write a program to enter the values of two variables 'a' and 'b' from user and then check if both the conditions 'a < 50' and 'a < b' are true.

**CODE:**

```
#include <iostream>
using namespace std;
int main(){
    int a,b;
    bool result;
    cout<<"Enter two number: ";
    cin>>a>>b;
    result= (a<50 && a<b);
    cout<<result;
    return 0;
}
```

**OUTPUT:**

```
C:\Users\Vishal\Desktop
Enter two number: 45 35
0
-----
Process exited after 6.632 seconds with return value 0
Press any key to continue . . . |
```

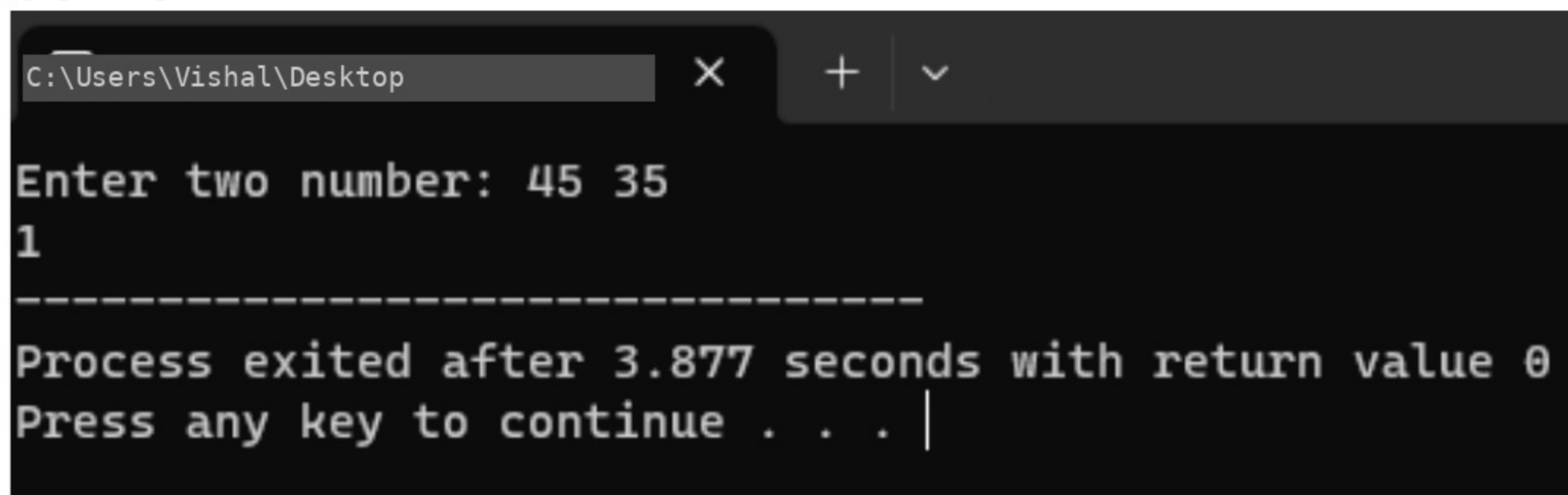


**QUESTION 9:**

Write a program to enter the values of two variables 'a' and 'b' from user and then check if atleast one of the conditions 'a < 50' and 'a < b' is true.

**CODE:**

```
#include <iostream>
using namespace std;
int main(){
    int a,b;
    bool result;
    cout<<"Enter two number: ";
    cin>>a>>b;
    result= (a<50 || a<b);
    cout<<result;
    return 0;
}
```

**OUTPUT:**

```
C:\Users\Vishal\Desktop  X  +  v

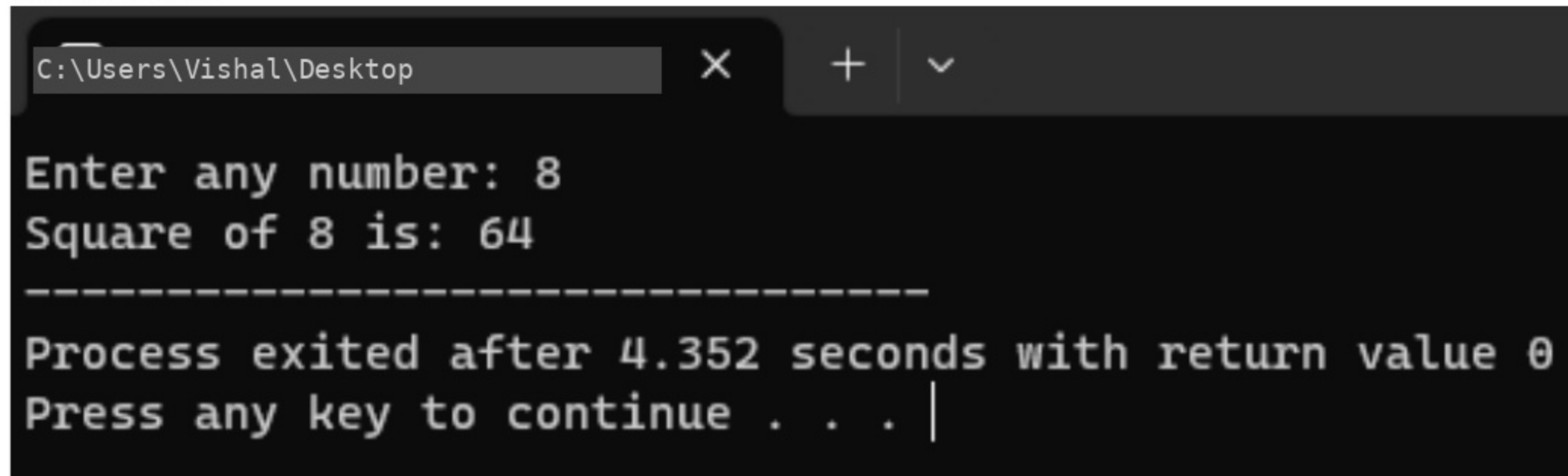
Enter two number: 45 35
1
-----
Process exited after 3.877 seconds with return value 0
Press any key to continue . . . |
```

**QUESTION 10:**

Implement a program that calculates the square of a number entered by the user (using assignment operator).

**CODE:**

```
#include <iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter any number: ";
    cin>>num;
    cout<<"Square of "<<num<<" is: ";
    num*=num;
    cout<<num;
    return 0;
}
```

**OUTPUT:**

```
C:\Users\Vishal\Desktop  X  +  v

Enter any number: 8
Square of 8 is: 64
-----
Process exited after 4.352 seconds with return value 0
Press any key to continue . . . |
```